

6. Resets

6.1 Overview

This MCU provides the following 21 types of reset.

It also has a function to control the reset generation by only secure program access.

System reset is all factors such that can initialize CPU core. All reset factors except VBATT_POR are system reset for this product.

On the other hand, CPU has a reset that initializes itself individually. The reset is not a system reset. See [section 2, CPU](#).

[Table 6.1](#) lists the reset names and sources.

Table 6.1 Reset names and sources

Reset Name	Source
System reset	RES Pin reset
	Voltage input to the RES pin is driven low
	Power-on reset (POR)
	VCC falls (voltage detection: V_{POR})* ¹
	Voltage Monitor 0 reset
	VCC falls (voltage detection: V_{det0})* ¹
	Voltage Monitor 1 reset
	VCC falls (voltage detection: V_{det1})* ¹ VCC rises (voltage detection: V_{det1})* ¹
	Voltage Monitor 2 reset
	VCC falls (voltage detection: V_{det2})* ¹ VCC rises (voltage detection: V_{det2})* ¹
	Voltage Monitor 4 reset
	VCC falls (voltage detection: V_{det4})* ¹ VCC rises (voltage detection: V_{det4})* ¹
	Voltage Monitor 5 reset
	VCC falls (voltage detection: V_{det5})* ¹ VCC rises (voltage detection: V_{det5})* ¹
	Core Voltage Monitor reset
	VDD falls (voltage detection: V_{det_VDDL})* ¹ VDD rises (voltage detection: V_{det_VDDH})* ¹
	Temperature Monitor reset
	Detect temperature outside a certain range
	Independent Watchdog Timer reset
	IWDT underflow or refresh error
CPU reset	CPU0 Watchdog Timer reset
	WDT underflow or refresh error for CPU0
	CPU1 Watchdog Timer reset
	WDT underflow or refresh error for CPU1
	CPU0 Lockup reset
	This reset is generated when CPU0 encounters lockup
	CPU1 Lockup reset
	This reset is generated when CPU1 encounters lockup
	Bus Error reset
	BUS error (MSAU error, MPU error, Illegal address error, TZF error, Slave Bus error, Bufferable write error)
Memory Error reset	Common Memory Error reset
	RAM error (ECC error of SRAM)
	Local Memory 0 Error reset
	RAM error (ECC error of CPU0 Cache or TCM)
	Local Memory 1 Error reset
	RAM error (ECC error of CPU1 Cache or TCM)
Deep Standby reset	Deep Software Standby reset
	Deep Software Standby mode is canceled by an interrupt
Software reset	Software reset
	Register setting (use the software reset bit AIRCR.SYSRESETREQ)
VBATT_POR reset	VBATT_R falls (voltage detection: $V_{PORBATT}$)* ¹

Note 1. For details on the voltages to be monitored (V_{POR} , V_{det0} , V_{det1} , V_{det2} , V_{det4} , V_{det5} , V_{det_VDDL} , V_{det_VDDH} and $V_{PORBATT}$), see [section 8, Programmable Voltage Detection \(PVD\)](#), [section 12, Battery Backup Function](#), and [section 60, Electrical Characteristics](#).

The internal state and pins are initialized by a reset. [Table 6.2](#) and [Table 6.3](#) list the targets initialized by resets.

Table 6.2 Reset detect flags initialized by each reset source (1 of 4)

Flag to be initialized	Reset source										
	RES pin reset	Power-on reset	Voltage monitor 0 reset	Core voltage monitor reset	Voltage monitor 4 reset	Voltage monitor 5 reset	Temperature monitor reset	Independent watchdog timer reset	CPU0 watchdog timer reset	CPU1 watchdog timer reset	CPU0 lockup reset
Power-On Reset Detect Flag (RSTSR0.PORF)	✓	—	—	—	—	—	—	—	—	—	—
Voltage Monitor 0 Reset Detect Flag (RSTSR0.PVD0RF)	✓	✓	—	—	—	—	—	—	—	—	—
Core Voltage Monitor Reset Detect Flag (RSTSR3.CVMRF)	✓	✓	✓	—	—	—	—	—	—	—	—
Temperature Monitor Reset Detect Flag (RSTSR3.TEMPRF)	✓	✓	✓	—	—	—	—	—	—	—	—
Independent Watchdog Timer Reset Detect Flag (RSTSR1.IWDTRF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
CPU0 Watchdog Timer Reset Detect Flag (RSTSR1.WDT0RF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
CPU1 Watchdog Timer Reset Detect Flag (RSTSR1.WDT1RF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
CPU0 Lockup Reset Detect Flag (RSTSR1.CLU0RF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
CPU1 Lockup Reset Detect Flag (RSTSR1.CLU1RF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
Voltage Monitor 1 Reset Detect Flag (RSTSR0.PVD1RF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
Voltage Monitor 2 Reset Detect Flag (RSTSR0.PVD2RF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
Voltage Monitor 4 Reset Detect Flag (RSTSR0.PVD4RF)	✓	✓	✓	—	—	—	—	—	—	—	—
Voltage Monitor 5 Reset Detect Flag (RSTSR0.PVD5RF)	✓	✓	✓	—	—	—	—	—	—	—	—
Software Reset Detect Flag (RSTSR1.SWRF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
Bus error Reset Detect Flag (RSTSR1.BUSRF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
Common Memory Error Reset Detect Flag (RSTSR1.CMRF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—

Table 6.2 Reset detect flags initialized by each reset source (2 of 4)

Flag to be initialized	Reset source										
	RES pin reset	Power-on reset	Voltage monitor 0 reset	Core voltage monitor reset	Voltage monitor 4 reset	Voltage monitor 5 reset	Temperature monitor reset	Independent watchdog timer reset	CPU0 watchdog timer reset	CPU1 watchdog timer reset	CPU0 lockup reset
Local Memory 0 Error Reset Detect Flag (RSTSR1.LM0RF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
Local Memory 1 Error Reset Detect Flag (RSTSR1.LM1RF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
Deep Software Standby Reset Detect Flag (RSTSR0.DPSRSTF)	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
Cold Start/Warm Start Determination Flag (RSTSR2.CWSF)	—	✓	—	—	—	—	—	—	—	—	—

Table 6.2 Reset detect flags initialized by each reset source (3 of 4)

Flag to be initialized	Reset source											
	CPU1 Lockup reset	Voltage monitor 1 reset	Voltage monitor 2 reset	Software reset	Bus error reset	Common memory error reset	Local memory 0 error reset	Local memory 1 error reset	Deep software standby reset			VBATT_POR*1
									Deep software standby mode 1	Deep software standby mode 2	Deep software standby mode 3	
Power-On Reset Detect Flag (RSTSR0.PORF)	—	—	—	—	—	—	—	—	—	—	—	—
Voltage Monitor 0 Reset Detect Flag (RSTSR0.PVD0RF)	—	—	—	—	—	—	—	—	—	—	—	—
Core Voltage Monitor Reset Detect Flag (RSTSR3.CVMRF)	—	—	—	—	—	—	—	—	—	—	—	—
Temperature Monitor Reset Detect Flag (RSTSR3.TEMPRF)	—	—	—	—	—	—	—	—	—	—	—	—
Independent Watchdog Timer Reset Detect Flag (RSTSR1.IWDTRF)	—	—	—	—	—	—	—	—	—	✓	✓	—
CPU0 Watchdog Timer Reset Detect Flag (RSTSR1.WDT0RF)	—	—	—	—	—	—	—	—	—	✓	✓	—
CPU1 Watchdog Timer Reset Detect Flag (RSTSR1.WDT1RF)	—	—	—	—	—	—	—	—	—	✓	✓	—

Table 6.2 Reset detect flags initialized by each reset source (4 of 4)

Flag to be initialized	Reset source											
	CPU1 Lockup reset	Voltage monitor 1 reset	Voltage monitor 2 reset	Software reset	Bus error reset	Common memory error reset	Local memory 0 error reset	Local memory 1 error reset	Deep software standby reset			VBATT_POR ^{*1}
									Deep software standby mode 1	Deep software standby mode 2	Deep software standby mode 3	
CPU0 Lockup Reset Detect Flag (RSTSR1.CLU0RF)	—	—	—	—	—	—	—	—	—	✓	✓	—
CPU1 Lockup Reset Detect Flag (RSTSR1.CLU1RF)	—	—	—	—	—	—	—	—	—	✓	✓	—
Voltage Monitor 1 Reset Detect Flag (RSTSR0.PVD1RF)	—	—	—	—	—	—	—	—	—	—	—	—
Voltage Monitor 2 Reset Detect Flag (RSTSR0.PVD2RF)	—	—	—	—	—	—	—	—	—	—	—	—
Voltage Monitor 4 Reset Detect Flag (RSTSR0.PVD4RF)	—	—	—	—	—	—	—	—	—	—	—	—
Voltage Monitor 5 Reset Detect Flag (RSTSR0.PVD5RF)	—	—	—	—	—	—	—	—	—	—	—	—
Software Reset Detect Flag (RSTSR1.SWRF)	—	—	—	—	—	—	—	—	—	✓	✓	—
Bus error Reset Detect Flag (RSTSR1.BUSRF)	—	—	—	—	—	—	—	—	—	✓	✓	—
Common Memory Error Reset Detect Flag (RSTSR1.CMRF)	—	—	—	—	—	—	—	—	—	✓	✓	—
Local Memory 0 Error Reset Detect Flag (RSTSR1.LM0RF)	—	—	—	—	—	—	—	—	—	✓	✓	—
Local Memory 1 Error Reset Detect Flag (RSTSR1.LM1RF)	—	—	—	—	—	—	—	—	—	✓	✓	—
Deep Software Standby Reset Detect Flag (RSTSR0.DPSRSTF)	—	—	—	—	—	—	—	—	—	—	—	—
Cold Start/Warm Start Determination Flag (RSTSR2.CWSF)	—	—	—	—	—	—	—	—	—	—	—	—

Note: ✓ : Initialized to 0

— : Not initialized

Note 1. For VBATT_POR details, see [section 12, Battery Backup Function](#).

Table 6.3 Module-related registers initialized by each reset source (1 of 7)

Registers to be initialized		Reset source										
		RES# pin reset	Power-on reset	Voltage monitoring 0 reset	Core Voltage monitor reset	Voltage monitor 4 reset	Voltage monitor 5 reset	Temperature monitor reset	Independent watchdog timer reset	CPU0 watchdog timer reset	CPU1 watchdog timer reset	CPU0 lockup reset
Voltage Monitor Function 1 registers	PVD1CR0, PVD1CMPCR, PVD1FCR	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—
	PVD1CR1, PVD1SR	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—
Voltage Monitor Function 2 registers	PVD2CR0, PVD2CMPCR, PVD2FCR	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—
	PVD2CR1, PVD2SR	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—
Voltage Monitor Function 4 registers	PVD4CR0, PVD4CMPCR, PVD4FCR	✓	✓	✓	—	—	—	—	—	—	—	—
	PVD4CR1, PVD4SR	✓	✓	✓	—	—	—	—	—	—	—	—
Voltage Monitor Function 5 registers	PVD5CR0, PVD5CMPCR, PVD5FCR	✓	✓	✓	—	—	—	—	—	—	—	—
	PVD5CR1, PVD5SR	✓	✓	✓	—	—	—	—	—	—	—	—
Temperature Reset Control Register	TEMPRCR	✓	✓	✓	—	—	—	—	—	—	—	—
SOSC registers	SOSCCR, SOSTDR, SOSTDSR, SOMCR	—	—	—	—	—	—	—	—	—	—	—
LOCO registers	LOCOUTCR	—	✓	✓	✓	✓	✓	✓	—	—	—	—
MOSC registers	MOMCR	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
HOCO registers	HOCOCR2	—	✓	✓	✓	✓	✓	✓	—	—	—	—
Pin states (except XCIN/XCOUT)		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Pin states (XCIN/XCOUT)		—	—	—	—	—	—	—	—	—	—	—
IO capture and tamper detection such as VBAT (sampling timing for RTC) (RTCIC0-2)	VBTICTLR, VBTICTLR2, VBTADSR, VBTADCR1, VBTADCR2, VBTADCR3, VBTNCWCR	—	—	—	—	—	—	—	—	—	—	—

Table 6.3 Module-related registers initialized by each reset source (2 of 7)

Registers to be initialized		Reset source										
		RES# pin reset	Power-on reset	Voltage monitoring 0 reset	Core Voltage monitor reset	Voltage monitor 4 reset	Voltage monitor 5 reset	Temperature monitor reset	Independent watchdog timer reset	CPU0 watchdog timer reset	CPU1 watchdog timer reset	CPU0 lockup reset
VBATT Battery Power Supply Switch Control Register 1	VBTBPCR1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
VBATT Battery Power Supply Switch Control Register 2	VBTBPCR2	—	—	—	—	—	—	—	—	—	—	—
VBATT Backup Enable Register	VBTBEBR	—	✓	—	—	—	—	—	—	—	—	—
Battery Backup Register	VBTBKRn	—	—	—	—	—	—	—	—	—	—	—
Independent Watchdog Timer registers	IWDTRR, IWDTCR, IWDTSR, IWDTRCR, IWDTCSTPR, IWDTS, IWDTPR	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Realtime Clock ^{*1} Register		—	—	—	—	—	—	—	—	—	—	—
Low-Power Timer registers	ULPTCNT, ULPTCMA, ULPTCMB, ULPTCR, ULPTMR1, ULPTMR2, ULPTMR3, ULPTIOC, ULPTISR, ULPTCMSR	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
USBFS registers	DPUSR0R, DPUSR1R	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Reset Flag	BUSERRST AT<Master Name>, MBWERRST AT, SRAMESR, SRAMEARn m (n = 0 to 3, m = 0 to 1), CCAEDST	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Reset Flag	See Table 6.2											

Table 6.3 Module-related registers initialized by each reset source (3 of 7)

Registers to be initialized		Reset source										
		RES# pin reset	Power-on reset	Voltage monitoring 0 reset	Core Voltage monitor reset	Voltage monitor 4 reset	Voltage monitor 5 reset	Temperature monitor reset	Independent watchdog timer reset	CPU0 watchdog timer reset	CPU1 watchdog timer reset	CPU0 lockup reset
Low-Power Function registers	DPSBYCR, DPSWCR, DPSIER0 to DPSIER5, DPSIFR0 to DPSIFR5, DPSIEGR0 to DPSIEGR4 LPSCR	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Low-Power Function registers	SYOCDRCR	—	✓	✓	✓	✓	✓	✓	—	—	—	—
System Reset Mask Control Register	SYRSTMSK 0, SYRSTMSK 1, SYRSTMSK 2	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
ARM Debug function	MCUSTAT, MCUCTRL, JBMDR, DBGSTR, DBGSTOPCR, DBGAUTH0, CACHEDBGCR, TRPORTCR, ALCTRL, TRPORTSZ, DBGVMCR, Cortex-M85 registers ^{*4} , Cortex-M33 Registers ^{*5} , CoreSight Registers ^{*6}	—	✓	✓	✓	✓	✓	✓	—	—	—	—
TRCLK Control registers	TRCKCR	—	✓	✓	✓	✓	✓	✓	—	—	—	—
Power Gating Control registers	PDRAMSCR 0, PDRAMSCR 1, PGSCR, PSSTCRn (n = 2 to 5)	—	✓	✓	✓	✓	✓	✓	—	—	—	—
Other than the above		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 6.3 Module-related registers initialized by each reset source (4 of 7)

Registers to be initialized		Reset source											
		CPU1 Lockup reset	Voltage monitoring 1 reset	Voltage monitoring 2 reset	Software reset	Bus error reset	Common memory error reset	Local memory 0 error reset	Local memory 1 error reset	Deep software standby reset			VBATT -selected voltage power-on reset
										Deep software standby mode 1 reset	Deep software standby mode 2 reset	Deep software standby mode 3 reset	
Voltage Monitor Function 1 registers	PVD1CR0, PVD1CMPCR, PVD1FCR	—	—	—	—	—	—	—	—	—	—	—	—
	PVD1CR1, PVD1SR	—	—	—	—	—	—	—	—	✓	✓	✓	—
Voltage Monitor Function 2 registers	PVD2CR0, PVD2CMPCR, PVD2FCR	—	—	—	—	—	—	—	—	—	—	—	—
	PVD2CR1, PVD2SR	—	—	—	—	—	—	—	—	✓	✓	✓	—
Voltage Monitor Function 4 registers	PVD4CR0, PVD4CMPCR, PVD4FCR	—	—	—	—	—	—	—	—	—	—	—	—
	PVD4CR1, PVD4SR	—	—	—	—	—	—	—	—	✓	✓	✓	—
Voltage Monitor Function 5 registers	PVD5CR0, PVD5CMPCR, PVD5FCR	—	—	—	—	—	—	—	—	—	—	—	—
	PVD5CR1, PVD5SR	—	—	—	—	—	—	—	—	✓	✓	✓	—
Temperature Reset Control Register	TEMPRCR	—	—	—	—	—	—	—	—	—	—	—	—
SOSC registers	SOSCCR, SOSTDR, SOSTDSR, SOMCR	—	—	—	—	—	—	—	—	—	—	—	✓
LOCO registers	LOCOUTCR	—	—	—	—	—	—	—	—	—	✓	✓	—
MOSC registers	MOMCR	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
HOCO registers	HOCOCR2	—	—	—	—	—	—	—	—	✓ ^{*3}	✓ ^{*3}	✓ ^{*3}	—
Pin states (except XCIN/XCOUT)		✓	✓	✓	✓	✓	✓	✓	✓	^{*2}	^{*2}	^{*2}	—
Pin states (XCIN/XCOUT)		—	—	—	—	—	—	—	—	—	—	—	✓

Table 6.3 Module-related registers initialized by each reset source (5 of 7)

Registers to be initialized		Reset source											
		CPU1 Lockup reset	Voltage monitoring 1 reset	Voltage monitoring 2 reset	Software reset	Bus error reset	Common memory error reset	Local memory 0 error reset	Local memory 1 error reset	Deep software standby reset			VBATT -selected voltage power-on reset
										Deep software standby mode 1 reset	Deep software standby mode 2 reset	Deep software standby mode 3 reset	
IO capture and tamper detection such as VBAT (sampling timing for RTC) (RTCIC0-2)	VBTICTLR, VBTICTLR2, VBTADSR, VBTADCR1, VBTADCR2, VBTADCR3, VBTNCWCR	—	—	—	—	—	—	—	—	—	—	—	✓
VBATT Battery Power Supply Switch Control Register 1	VBTBPCR1	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
VBATT Battery Power Supply Switch Control Register 2	VBTBPCR2	—	—	—	—	—	—	—	—	—	—	—	✓
VBATT Backup Enable Register	VBTBER	—	—	—	—	—	—	—	—	—	—	—	—
Battery Backup Register	VBTBKRN	—	—	—	—	—	—	—	—	—	—	—	✓
Independent Watchdog Timer registers	IWDTRR, IWDTCR, IWDTSR, IWDTRCR, IWDTCSTPR, IWDTTS, IWDTPR	✓	✓	✓	✓	✓	✓	✓	✓	—	✓	✓	—
Realtime Clock*1 Register		—	—	—	—	—	—	—	—	—	—	—	—

Table 6.3 Module-related registers initialized by each reset source (6 of 7)

Registers to be initialized		Reset source											
		CPU1 Lockup reset	Voltage monitoring 1 reset	Voltage monitoring 2 reset	Software reset	Bus error reset	Common memory error reset	Local memory 0 error reset	Local memory 1 error reset	Deep software standby reset			VBATT -selected voltage power-on reset
										Deep software standby mode 1 reset	Deep software standby mode 2 reset	Deep software standby mode 3 reset	
Low-Power Timer registers	ULPTCNT, ULPTCMA, ULPTCMB, ULPTCR, ULPTMR1, ULPTMR2, ULPTMR3, ULPTIOC, ULPTISR, ULPTCMSR	✓	✓	✓	✓	✓	✓	✓	✓	—	✓	✓	—
USBFS registers	DPUSR0R, DPUSR1R	✓	✓	✓	✓	✓	✓	✓	✓	—	✓	✓	—
Reset Flag	BUSERRSTAT<Master Name>, MBWERRSTAT, SRAMESR, SRAMEARNm (n = 0 to 3, m = 0 to 1), CCAEDST	✓	✓	✓	✓	—	—	—	—	✓	✓	✓	—
Reset Flag		See Table 6.2 .											
Low-Power Function registers	DPSBYCR, DPSWCR, DPSIER0 to DPSIER5, DPSIFR0 to DPSIFR5, DPSIEGR0 to DPSIEGR4, LPSCR	✓	✓	✓	✓	✓	✓	✓	✓	—	—	—	—
Low-Power Function registers	SYOCDRCR	—	—	—	—	—	—	—	—	—	—	—	—
System Reset Mask Control Register	SYRSTMSK0, SYRSTMSK1, SYRSTMSK2	—	—	—	—	—	—	—	—	—	—	—	—

Table 6.3 Module-related registers initialized by each reset source (7 of 7)

Registers to be initialized		Reset source											
		CPU1 Lockup reset	Voltage monitoring 1 reset	Voltage monitoring 2 reset	Software reset	Bus error reset	Common memory error reset	Local memory 0 error reset	Local memory 1 error reset	Deep software standby reset			VBATT -selected voltage power-on reset
										Deep software standby mode 1 reset	Deep software standby mode 2 reset	Deep software standby mode 3 reset	
ARM Debug function	MCUSTAT, MCUCTRL, JBMDR, DBGSTR, DBGSTOP CR, DBGAUTH0, CACHEDBGCR, TRPORTCR, ALCTRL, TRPORTSZ, DBGNVMCR, Cortex-M85 registers* ⁴ , Cortex-M33 registers* ⁵ , CoreSight registers* ⁶	—	—	—	—	—	—	—	—	✓* ³	✓* ³	✓* ³	—
TRCLK Control registers	TRCKCR	—	—	—	—	—	—	—	—	✓* ³	✓* ³	✓* ³	—
Power Gating Control registers	PDRAMSC R0, PDRAMSC R1, PGSCR, PSSTCRn (n = 2 to 5)	—	—	—	—	—	—	—	—	✓* ³	✓* ³	✓* ³	—
Other than the above		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—

Note: ✓ : Initialized
 — : Not initialized

Note 1. RTC has Software reset for RTC. Some control bits are not initialized by all types of resets. For details on the target bits, see [section 9, Clock Generation Circuit](#).

Note 2. Depend on setting of DPSBYCR.IOKEEP.

Note 3. After the debugger authentication passes, setting SYOCDRCR.DBGEN to 1 does not initialize this register or the function. For details, see [section 2, CPU](#).

Note 4. Cortex-M85 registers initialized on cold reset.

Note 5. Cortex-M33 registers initialized on cold reset.

Note 6. CoreSight SoC-400 TPIU registers, CoreSight SoC-400 ATB Funnel registers, CoreSight SoC-400 ATB Replicator registers, CoreSight SoC-400 CTI Replicator registers, CoreSight SoC-400 AHB-AP registers, CoreSight SoC-400 APB-AP registers, CoreSight SoC-400 Debug Port registers, CoreSight SoC-400 Timestamp Generator registers, CoreSight Trace Memory Controller registers.

The RTC is not initialized by any reset source. SOSC and LOCO can be selected as the clock sources of the RTC.

[Table 6.4](#) and [Table 6.5](#) show the states of SOSC and LOCO when a reset occurs.

Table 6.4 States of SOSC when a reset occurs

		Reset source	
		VBAT_POR reset	Other
SOSC	Enable or disable	Initialized to enable	Continue with the state that was selected before the reset occurred
	Drive capability	Initialized to standard drive capability	

Table 6.5 States of LOCO when a reset occurs

		Reset source	
		Power-on reset, Voltage-Monitor 0, 4, 5 reset, Core Voltage-Monitor reset, Temperature-Monitor reset and Deep software standby 2, 3 reset	Other
LOCO	Enable or disable	Initialized to enable	Continue with the accuracy that was trimmed by LOCOUTCR
	Oscillation accuracy ^{*1}	Initialized to accuracy before trimming by LOCOUTCR (accuracy: $\pm 15\%$)	

Note 1. If the LOCO that is trimmed by LOCOUTCR is selected as the RTC source clock, please be careful that LOCO oscillation accuracy will be initialized by Power-on reset, Voltage-Monitor 0, 4, 5 reset, Core Voltage-Monitor reset, Temperature-Monitor reset and Deep software standby 2, 3 reset.

When a reset is released, reset exception handling starts. For details on the reset exception handling, see [section 6.3.14. Determination of Reset Generation Source](#).

[Table 6.6](#) lists the pin related to the reset function.

Table 6.6 Pin related to reset

Pin name	I/O	Function
RES	Input	Reset pin

6.2 Register Descriptions

6.2.1 RSTSAR : Reset Security Attribution Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x3C4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	NONS EC3	NONS EC2	NONS EC1	NONS EC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NONSEC0	Non-secure Attribute bit 0 Target register: Reset Status Register 0 0: Secure 1: Non-secure	R/W
1	NONSEC1	Non-secure Attribute bit 1 Target register: Reset Status Register 1 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
2	NONSEC2	Non-secure Attribute bit 2 Target register: Reset Status Register 2 0: Secure 1: Non-secure	R/W
3	NONSEC3	Non-secure Attribute bit 3 Target register: Reset Status Register 3 0: Secure 1: Non-secure	R/W
31:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

NONSEC0 bit (Non-secure Attribute bit 0)

This bit controls the security attribute of RSTSR0.

NONSEC1 bit (Non-secure Attribute bit 1)

This bit controls the security attribute of RSTSR1.

NONSEC2 bit (Non-secure Attribute bit 2)

This bit controls the security attribute of RSTSR2.

NONSEC3 bit (Non-secure Attribute bit 3)

This bit controls the security attribute of RSTSR3.

6.2.2 RSTSR0 : Reset Status Register 0

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA40

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DPSR STF	PVD5 RF	PVD4 RF	—	PVD2 RF	PVD1 RF	PVD0 RF	PORF
Value after reset:	x ^{*1}	x ^{*1}	x ^{*1}	0	x ^{*1}	x ^{*1}	x ^{*1}	x ^{*1}

Bit	Symbol	Function	R/W
0	PORF	Power-On Reset Detect Flag 0: Power-on reset not detected 1: Power-on reset detected	R/W ^{*2}
1	PVD0RF	Voltage Monitor 0 Reset Detect Flag 0: Voltage monitor 0 reset not detected 1: Voltage monitor 0 reset detected	R/W ^{*2}
2	PVD1RF	Voltage Monitor 1 Reset Detect Flag 0: Voltage monitor 1 reset not detected 1: Voltage monitor 1 reset detected	R/W ^{*2}
3	PVD2RF	Voltage Monitor 2 Reset Detect Flag 0: Voltage monitor 2 reset not detected 1: Voltage monitor 2 reset detected	R/W ^{*2}
4	—	This bit is read as 0. The write value should be 0.	R/W
5	PVD4RF	Voltage Monitor 4 Reset Detect Flag 0: Voltage monitor 4 reset not detected 1: Voltage monitor 4 reset detected	R/W ^{*2}
6	PVD5RF	Voltage Monitor 5 Reset Detect Flag 0: Voltage monitor 5 reset not detected 1: Voltage monitor 5 reset detected	R/W ^{*2}

Bit	Symbol	Function	R/W
7	DPSRSTF	Deep Software Standby Reset Flag 0: Deep Software Standby Mode cancellation not requested by an interrupt or a reset*3. 1: Deep Software Standby Mode cancellation requested by an interrupt or a reset*3.	R/W*2

Note: S-TYPE-3, P-TYPE-2

Note 1. The value after reset depends on the reset source.

Note 2. Only 0 can be written to clear the flag. The flag must be cleared by writing 0 after 1 is read.

Note 3. Independent watchdog timer reset, Voltage Monitor 1 reset, Voltage Monitor 2 reset.

PORF flag (Power-On Reset Detect Flag)

The PORF flag indicates that a power-on reset occurred.

[Setting condition]

- When a power-on reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs
- When 0 is written after 1 is read from PORF flag.

PVD0RF flag (Voltage Monitor 0 Reset Detect Flag)

The PVD0RF flag indicates that the voltage monitor 0 reset has occurred.

[Setting condition]

- When a voltage monitor 0 reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from PVD0RF flag.

PVD1RF flag (Voltage Monitor 1 Reset Detect Flag)

The PVD1RF flag indicates that the voltage monitor 1 reset has occurred.

[Setting condition]

- When a voltage monitor 1 reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from PVD1RF flag.

PVD2RF flag (Voltage Monitor 2 Reset Detect Flag)

The PVD2RF flag indicates that the voltage monitor 2 reset has occurred.

[Setting condition]

- When a voltage monitor 2 reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from PVD2RF flag.

PVD4RF flag (Voltage Monitor 4 Reset Detect Flag)

The PVD4RF flag indicates that the voltage monitor 4 reset has occurred.

[Setting condition]

- When a voltage monitor 4 reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from PVD4RF flag.

PVD5RF flag (Voltage Monitor 5 Reset Detect Flag)

The PVD5RF flag indicates that the voltage monitor 5 reset has occurred.

[Setting condition]

- When a voltage monitor 5 reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from PVD5RF flag.

DPSRSTF flag (Deep Software Standby Reset Flag)

The DPSRSTF flag indicates that Deep Software Standby Mode is canceled by an external or internal interrupt, and that an internal reset (Deep Software Standby reset) occurs when an exception from Deep Software Standby Mode occurs.

[Setting condition]

- When Deep Software Standby Mode is cancelled by an external or an internal interrupt. For details, see [section 11, Low Power Mode](#).

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from DPSRSTF flag.

6.2.3 RSTSR1 : Reset Status Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0C0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	LM1R F	CLU1 RF	—	—	WDT1 RF	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	x*1	x*1	0	0	x*1	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	CMRF	—	—	—	BUSR F	—	—	—	—	LM0R F	CLU0 RF	—	SWRF	WDT0 RF	IWDT RF
Value after reset:	0	x*1	0	0	0	x*1	0	0	0	0	x*1	x*1	0	x*1	x*1	x*1

Bit	Symbol	Function	R/W
0	IWDTRF	Independent Watchdog Timer Reset Detect Flag 0: Independent watchdog timer reset not detected 1: Independent watchdog timer reset detected	R/W ²
1	WDT0RF	Watchdog Timer 0 Reset Detect Flag 0: Watchdog timer 0 reset not detected 1: Watchdog timer 0 reset detected	R/W ²
2	SWRF	Software Reset Detect Flag 0: Software reset not detected 1: Software reset detected	R/W ²
3	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
4	CLU0RF	CPU0 Lockup Reset Detect Flag 0: CPU0 lockup reset not detected 1: CPU0 lockup reset detected	R/W
5	LM0RF	Local memory 0 error Reset Detect Flag 0: Local memory 0 error reset not detected 1: Local memory 0 error reset detected	R/W ²
9:6	—	These bits are read as 0. The write value should be 0.	R/W
10	BUSRF	Bus Error Reset Detect Flag 0: Bus error reset not detected 1: Bus error reset detected	R/W ²
13:11	—	These bits are read as 0. The write value should be 0.	R/W
14	CMRF	Common Memory Error Reset Detect Flag 0: Common memory error reset not detected 1: Common memory error reset detected	R/W ²
16:15	—	These bits are read as 0. The write value should be 0.	R/W
17	WDT1RF	Watchdog Timer 1 Reset Detect Flag 0: Watchdog timer 1 reset not detected 1: Watchdog timer 1 reset detected	R/W ²
19:18	—	These bits are read as 0. The write value should be 0.	R/W
20	CLU1RF	CPU1 Lockup Reset Detect Flag 0: CPU1 lockup reset not detected 1: CPU1 lockup reset detected	R/W ²
21	LM1RF	Local memory 1 error Reset Detect Flag 0: Local memory 1 error reset not detected 1: Local memory 1 error reset detected	R/W ²
31:22	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note 1. The value after reset depends on the reset source.

Note 2. Only 0 can be written to clear the flag. The flag must be cleared by writing 0 after 1 is read.

IWDTRF flag (Independent Watchdog Timer Reset Detect Flag)

The IWDTRF flag indicates that an Independent watchdog timer reset occurred.

[Setting condition]

- When an independent watchdog timer reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from IWDTRF flag.

WDT0RF flag (Watchdog Timer 0 Reset Detect Flag)

The WDT0RF flag indicates that a CPU0 watchdog timer reset occurred.

[Setting condition]

- When a CPU0 watchdog timer reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from WDT0RF flag.

SWRF flag (Software Reset Detect Flag)

The SWRF flag indicates that a Software reset occurred.

[Setting condition]

- When a software reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from SWRF flag.

CLU0RF flag (CPU0 Lockup Reset Detect Flag)

The CLU0RF flag indicates that a CPU0 lockup reset occurred.

[Setting condition]

- When a CPU0 Lockup reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from CLU0RF flag.

LM0RF flag (Local memory 0 error Reset Detect Flag)

The LM0RF flag indicates that a Local memory 0 error reset (ECC error of CPU0 Cache or TCM) occurred.

[Setting condition]

- When a Local memory 0 error reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from LM0RF flag.

BUSRF flag (Bus Error Reset Detect Flag)

The BUSRF flag indicates that BUS error reset (MSAU error, MMPU error, Illegal address error, TZF error, Slave Bus error, Bufferable write error) occurred.

[Setting condition]

- When a bus slave MPU error reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs
- When 0 is written after 1 is read from BUSRF flag.

There are several causes for setting this reset flag.

If you want to identify the reset factor, see [section 15, Buses](#).

CMRF flag (Common Memory Error Reset Detect Flag)

The CMRF flag indicates that a Common memory error reset (ECC error of SRAM) occurred.

[Setting condition]

- When a Common memory error reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from CMRF flag.

There are several causes for setting this reset flag.

If you want to identify the reset factor, see [section 57, SRAM](#).

WDT1RF flag (Watchdog Timer 1 Reset Detect Flag)

The WDT1RF flag indicates that a CPU1 watchdog timer reset occurred.

[Setting condition]

- When a CPU1 watchdog timer reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from WDT1RF flag.

CLU1RF flag (CPU1 Lockup Reset Detect Flag)

The CLU1RF flag indicates that a CPU1 lockup reset occurred.

[Setting condition]

- When a CPU1 lockup reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from CLU1RF flag.

LM1RF flag (Local memory 1 error Reset Detect Flag)

The LM1RF flag indicates that a Local memory 1 error reset (ECC error of CPU1 Cache or TCM) occurred.

[Setting condition]

- When a Local memory 1 error reset occurs.

[Clearing conditions]

- When a reset listed in [Table 6.2](#) occurs.
- When 0 is written after 1 is read from LM1RF flag.

6.2.4 RSTSR2 : Reset Status Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA44

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CWSF
Value after reset:	0	0	0	0	0	0	0	x*1

Bit	Symbol	Function	R/W
0	CWSF	Cold/Warm Start Determination Flag 0: Cold start 1: Warm start	R/W ²
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note 1. The value after reset depends on the reset source.

Note 2. Only 1 can be written to set the flag.

RSTSR2 determines whether a power-on reset caused the reset processing (cold start) or a reset signal input during operation caused the reset processing (warm start).

CWSF flag (Cold/Warm Start Determination Flag)

The CWSF flag indicates the type of reset processing, either cold start or warm start. CWSF flag is initialized by a power-on reset. It is not initialized by a reset signal generated by the RES pin.

[Setting condition]

- When 1 is written by software. Writing 0 to CWSF does not set it to 0.

[Clearing condition]

- When a reset listed in [Table 6.2](#) occurs.

6.2.5 RSTSR3 : Reset Status Register 3

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA48

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TEMP RF	—	—	—	—	—	—	CVMR F
Value after reset:	x*1	0	0	0	0	0	0	x*1

Bit	Symbol	Function	R/W
0	CVMRF	Core Voltage Monitor Reset Detect Flag 0: Core voltage monitor reset not detected 1: Core voltage monitor reset detected	R/W ²
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	TEMPRF	Temperature Monitor Reset Detect Flag 0: Temperature monitor reset not detected 1: Temperature monitor reset detected	R/W ²

Note: S-TYPE-3, P-TYPE-2

Note 1. The value after reset depends on the reset source.

Note 2. Only 0 can be written to clear the flag. The flag must be cleared by writing 0 after 1 is read.

CVMRF flag (Core Voltage Monitor Reset Detect Flag)

The CVMRF indicates that Core voltage fell below $V_{\text{det_VDDL}}$.

The CVMRF indicates that Core voltage rised above $V_{\text{det_VDDH}}$.

[Setting condition]

- When a Core voltage monitor reset occurs.

[Clearing condition]

- When a reset listed in [Table 6.3](#) occurs.
- When 0 is written after 1 is read from CVMRF flag.

TEMPRF flag (Temperature Monitor Reset Detect Flag)

The TEMPRF indicates that Temperature monitor reset has occurred.

[Setting condition]

- When a Temperature monitor reset occurs.

[Clearing condition]

- When a reset listed in [Table 6.3](#) occurs.
- When 0 is written after 1 is read from TEMPRF flag.

6.2.6 SYRSTMSK0 : System Reset Mask Control Register 0

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xAD0

Bit position:	7	6	5	4	3	2	1	0
Bit field:	BUSM ASK	CMM ASK	LM0M ASK	CLU0 MASK	—	SWMA SK	WDT0 MASK	IWDT MASK
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IWDTMASK	Independent Watchdog Timer Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
1	WDT0MASK	CPU0 Watchdog Timer Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
2	SWMASK	Software Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	CLU0MASK	CPU0 Lockup Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
5	LM0MASK	Local Memory 0 Error Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
6	CMMASK	Common Memory Error Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
7	BUSMASK	Bus Error Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W

Note: S-TYPE-6, P-TYPE-2

Note: Set the PRCR.PRC5 bit to 1 (write enabled) before rewriting this register.

The SYRSTMSK0 is a register that controls the occurrence of reset.

The IWDTMASK bit cannot be rewritten while the independent watchdog timer is operating. The WDT0MASK bit cannot be rewritten while the CPU0 watchdog timer is operating.

6.2.7 SYRSTMSK1 : System Reset Mask Control Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xAD4

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	LM1M ASK	CLU1 MASK	—	—	WDT1 MASK	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	WDT1MASK	CPU1 Watchdog Timer Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W

Bit	Symbol	Function	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	CLU1MASK	CPU1 Lockup Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
5	LM1MASK	Local Memory 1 Error Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6, P-TYPE-2

Note: Set the PRCR.PRC5 bit to 1 (write enabled) before rewriting this register.

The SYRSTMSK1 is a register that controls the occurrence of reset.

6.2.8 SYRSTMSK2 : System Reset Mask Control Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xAD8

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PVD2 MASK	PVD1 MASK
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PVD1MASK	Voltage Monitor 1 Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
1	PVD2MASK	Voltage Monitor 2 Reset Mask 0: Reset occurrence is enabled 1: Reset occurrence is disabled	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6, P-TYPE-2

Note: Set the PRCR.PRC5 bit to 1 (write enabled) before rewriting this register.

The SYRSTMSK2 is a register that controls the occurrence of reset.

6.2.9 TEMPRCR : Temperature Monitor Reset Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xADC

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	TSNK EEP	CMPE N	TSNE N	TEMP REN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TEMPREN	Temperature Reset Enable 0: Temperature reset is disabled 1: Temperature reset is enabled	R/W
1	TSNEN	Temperature Sensor Enable 0: Temperature sensor is disabled 1: Temperature sensor is enabled	R/W

Bit	Symbol	Function	R/W
2	COMPEN	Comparator Enable 0: Comparator is disabled 1: Comparator is enabled	R/W
3	TSNKEEP	Temperature Sensor Latch Control 0: Temperature sensor latch is opened 1: Temperature sensor latch is closed	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6, P-TYPE-2

Note: Set the PRCR.PRC5 bit to 1 (write enabled) before rewriting this register.

6.2.10 TEMPRLR : Temperature Monitor Reset Lock Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xAE0

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	LOCK
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	LOCK	Temperature Monitor Reset Control Register Lock 0: Temperature monitor reset control register is unlocked 1: Temperature monitor reset control register is locked	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6, P-TYPE-2

Note: Set the PRCR.PRC5 bit to 1 (write enabled) before rewriting this register.

Note: This register accepts write accesses only twice. If you write to TEMPRLR 3 times or more, please reset it and re-write.

6.3 Operation

6.3.1 RES Pin Reset

The RES pin generates this reset. When the RES pin is driven low, all the processing in progress is aborted and the MCU enters a reset state. To successfully reset the MCU, the RES pin must be held low for the power supply stabilization time specified at power-on.

When the RES pin is driven high from low, the internal reset is canceled after the post-RES cancellation wait time (t_{RESWT}) elapses. The CPU then starts the reset exception handling.

For details, see [section 60, Electrical Characteristics](#).

6.3.2 Power-On Reset

The power-on reset circuit generates this internal reset. If the RES pin is in a high-level state when power is supplied, a power-on reset is generated.

After VCC exceeds V_{POR} and the specified period (power-on reset time) elapses, the internal reset is canceled, and the CPU starts the reset exception handling. The power-on reset time is the stabilization period for the external power supply and the MCU circuit.

After a power-on reset is generated, the PORF flag in the RSTSR0 is set to 1. The PORF flag is initialized by the RES pin reset.

[Figure 6.1](#) shows examples of operations during a power-on reset and voltage monitor 0 reset.

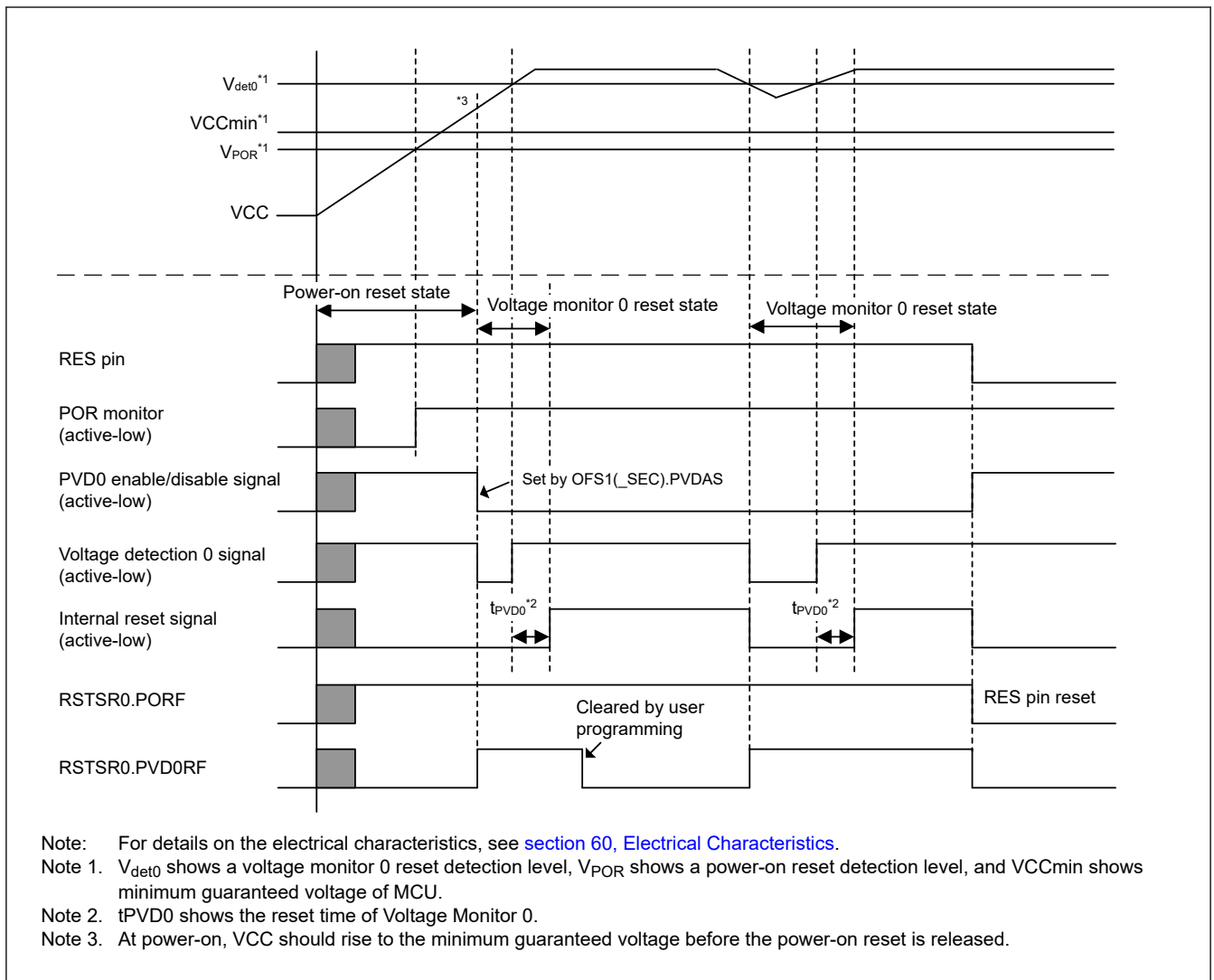


Figure 6.1 Example of operations during a power-on reset and Voltage Monitor 0 Reset

6.3.3 Voltage Monitor Reset

The voltage monitor 0 reset is an internal reset generated by the voltage monitor circuit. If the Voltage Detection 0 Circuit Start bit (PVDAS) in Option Function Select Register 1 (OFS1) is 0 (voltage monitor 0 reset is enabled after a reset) and VCC falls below V_{det0} , the RSTSR0.PVD0RF flag is set to 1 and the voltage detection circuit generates a voltage monitor 0 reset. Clear the OFS1.PVDAS bit to 0 if the voltage monitor 0 reset is to be used. After VCC exceeds V_{det0} and the voltage monitor 0 reset time (t_{PVD0}) elapses, the internal reset is canceled, and the CPU starts the reset exception handling. The V_{det0} voltage detection level can be changed by the setting in the VDSEL[1:0] bits in Option Function Select Register 1 (OFS1).

- Case of PVDiFCR.RHSEL = 0 and PVDjFCR.RHSEL = 0 (i = 1, 2, j = 4, 5)

When the Voltage Monitor 1 Interrupt/Reset Enable bit (RIE) is set to 1 (enabling generation of a reset or interrupt by the voltage detection circuit) and the Voltage Monitor 1 Circuit Mode Select bit (RI) is set to 1 (selecting generation of a reset in response to detection of a low voltage) in Voltage Monitor 1 Circuit Control Register 0 (PVD1CR0), the RSTSR0.PVD1RF flag is set to 1 and the voltage detection circuit generates a Voltage Monitor 1 reset if VCC falls to or below V_{det1} .

Timing for release from the Voltage Monitor 1 reset state is selectable in the Voltage Monitor 1 Reset Negate Select bit (RN) in PVD1CR0. When the RN bit is 0 and VCC falls to or below V_{det1} , the CPU is released from the internal reset state and starts reset exception handling when the PVD1 reset time (t_{PVD1}) elapses after VCC rises above V_{det1} . When the PVD1CR0.RN bit is 1 and VCC falls to or below V_{detm} , the CPU is released from the internal reset state and starts reset exception handling when the PVD1 reset time (t_{PVD1}) elapses.

Likewise, when the Voltage Monitor 2 Interrupt/Reset Enable bit (RIE) is set to 1 (enabling generation of a reset or interrupt by the voltage detection circuit) and the Voltage Monitor 2 Circuit Mode Select bit (RI) is set to 1 (selecting generation of a reset in response to detection of a low voltage) in Voltage Monitor 2 Circuit Control Register 0 (PVD2CR0), the RSTSR0.PVD2RF flag is set to 1 and the voltage detection circuit generates a Voltage Monitor 2 reset if VCC falls to or below V_{det2} .

Timing for release from the Voltage Monitor 2 reset state is selectable in the Voltage Monitor 2 Reset Negate Select bit (RN) in PVD2CR0. When the RN bit is 0 and VCC falls to or below V_{det2} , the CPU is released from the internal reset state and starts reset exception handling when the PVD2 reset time (t_{PVD2}) elapses after VCC rises above V_{det2} . When the PVD2CR0.RN bit is 1 and VCC falls to or below V_{detm} , the CPU is released from the internal reset state and starts reset exception handling when the PVD2 reset time (t_{PVD2}) elapses.

When the Voltage Monitor 4 Reset Enable bit (RE) is set to 1 (enabling generation of a reset by the voltage detection circuit) in Voltage Monitor 4 Circuit Control Register 0 (PVD4CR0), the RSTSR0.PVD4RF flag is set to 1 and the voltage detection circuit generates a Voltage Monitor 4 reset if VCC falls to or below V_{det4} .

The CPU is then released from the internal reset state and starts reset exception handling when the PVD4 reset time (t_{PVD4}) elapses after VCC rises above V_{det4} .

When the Voltage Monitor 5 Reset Enable bit (RE) is set to 1 (enabling generation of a reset by the voltage detection circuit) in Voltage Monitor 5 Circuit Control Register 0 (PVD5CR0), the RSTSR0.PVD5RF flag is set to 1 and the voltage detection circuit generates a Voltage Monitor 5 reset if VCC falls to or below V_{det5} .

The CPU is then released from the internal reset state and starts reset exception handling when the PVD5 reset time (t_{PVD5}) elapses after VCC rises above V_{det5} .

Figure 6.2 shows examples of operations during the voltage monitor 1 and 2 reset.

Figure 6.3 shows examples of operations during the voltage monitor 4 and 5 reset.

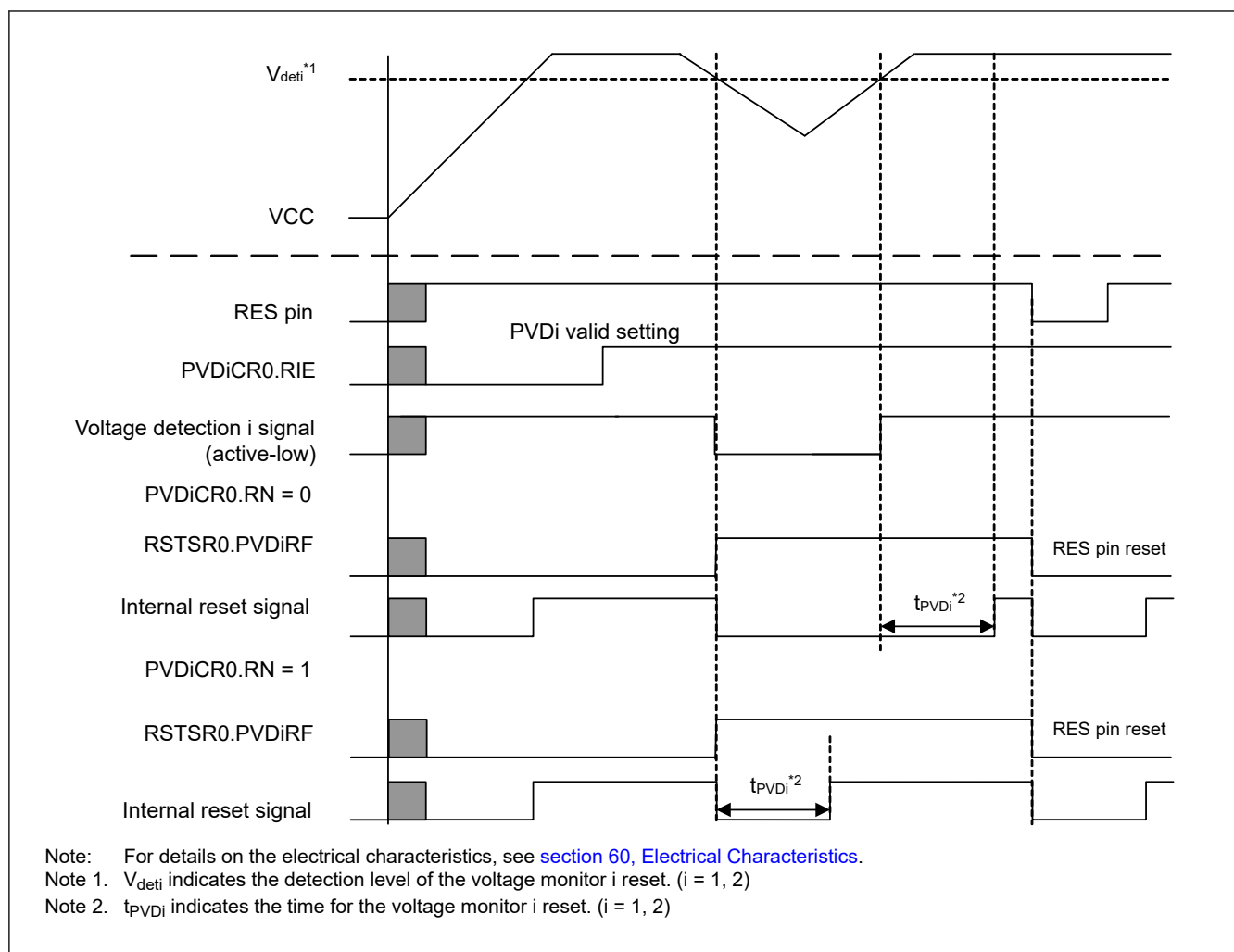


Figure 6.2 Example of operations during voltage monitor i reset ($i = 1, 2$)

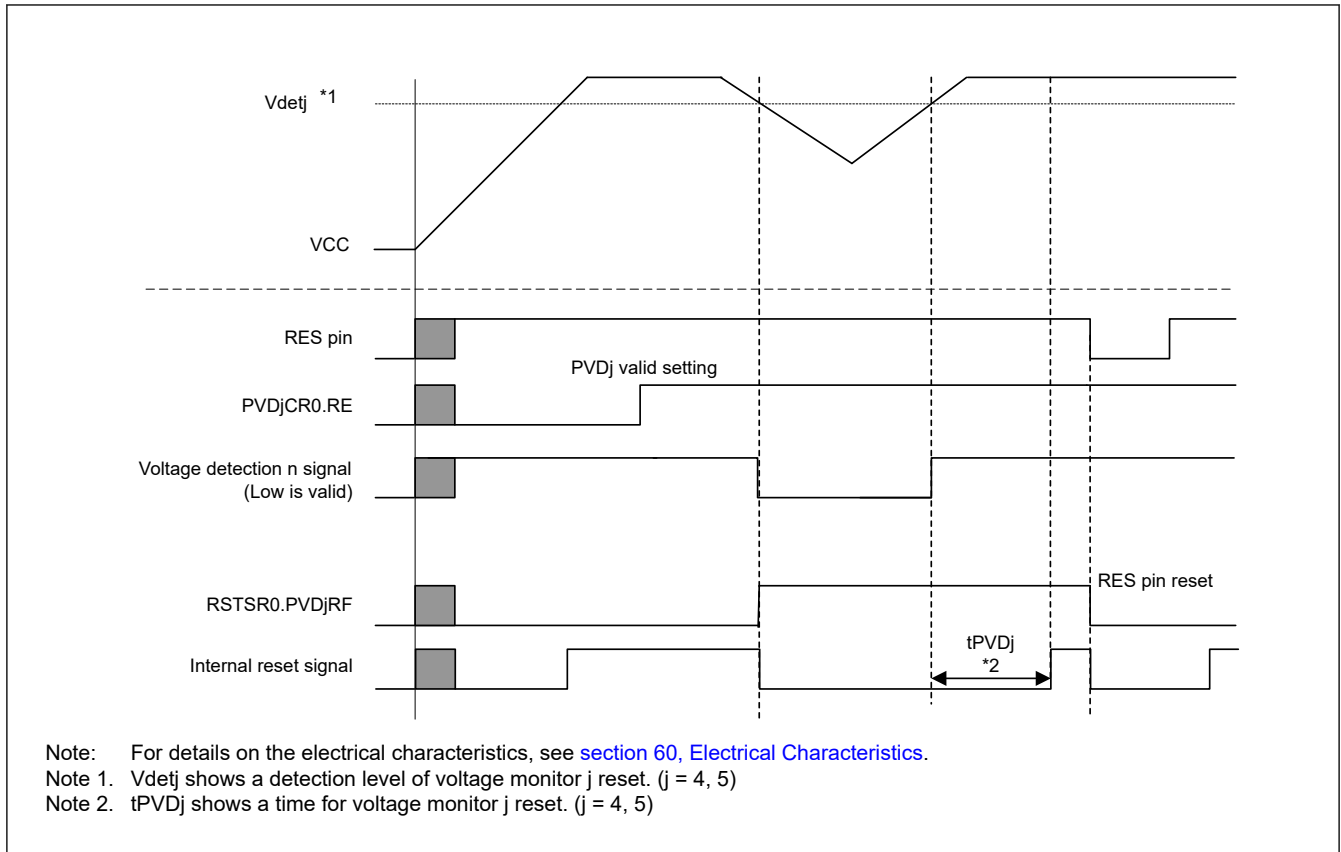


Figure 6.3 Example of operations during voltage monitor j reset ($j = 4, 5$)

- Case of $PVDiFCR.RHSEL = 1$ and $PVDjFCR.RHSEL = 1$ ($i = 1, 2, j = 4, 5$)

When the Voltage Monitor 1 Interrupt/Reset Enable bit (RIE) is set to 1 (enabling generation of a reset or interrupt by the voltage detection circuit) and the Voltage Monitor 1 Circuit Mode Select bit (RI) is set to 1 (selecting generation of a reset in response to detection of a low voltage) in Voltage Monitor 1 Circuit Control Register 0 (PVD1CR0), the RSTSR0.PVD1RF flag is set to 1 and the voltage detection circuit generates a Voltage Monitor 1 reset if VCC rises to or above V_{det1} .

The CPU is then released from the internal reset state and starts reset exception handling when the PVD1 reset time (t_{PVD1}) elapses after VCC falls below V_{det1} .

Likewise, when the Voltage Monitor 2 Interrupt/Reset Enable bit (RIE) is set to 1 (enabling generation of a reset or interrupt by the voltage detection circuit) and the Voltage Monitor 2 Circuit Mode Select bit (RI) is set to 1 (selecting generation of a reset in response to detection of a low voltage) in Voltage Monitor 2 Circuit Control Register 0 (PVD2CR0), the RSTSR0.PVD2RF flag is set to 1 and the voltage detection circuit generates a Voltage Monitor 2 reset if VCC rises to or above V_{det2} .

The CPU is then released from the internal reset state and starts reset exception handling when the PVD2 reset time (t_{PVD2}) elapses after VCC falls below V_{det2} .

When the Voltage Monitor 4 Reset Enable bit (RE) is set to 1 (enabling generation of a reset by the voltage detection circuit) in Voltage Monitor 4 Circuit Control Register 0 (PVD4CR0), the RSTSR0.PVD4RF flag is set to 1 and the voltage detection circuit generates a Voltage Monitor 4 reset if VCC rises to or above V_{det4} .

The CPU is then released from the internal reset state and starts reset exception handling when the PVD4 reset time (t_{PVD4}) elapses after VCC falls below V_{det4} .

When the Voltage Monitor 5 Reset Enable bit (RE) is set to 1 (enabling generation of a reset by the voltage detection circuit) in Voltage Monitor 5 Circuit Control Register 0 (PVD5CR0), the RSTSR0.PVD5RF flag is set to 1 and the voltage detection circuit generates a Voltage Monitor 5 reset if VCC rises to or above V_{det5} .

The CPU is then released from the internal reset state and starts reset exception handling when the PVD5 reset time (t_{PVD5}) elapses after VCC falls below V_{det5} .

Figure 6.4 shows examples of operations during the voltage monitor 1 and 2 reset.

Figure 6.5 shows examples of operations during the voltage monitor 4 and 5 reset.

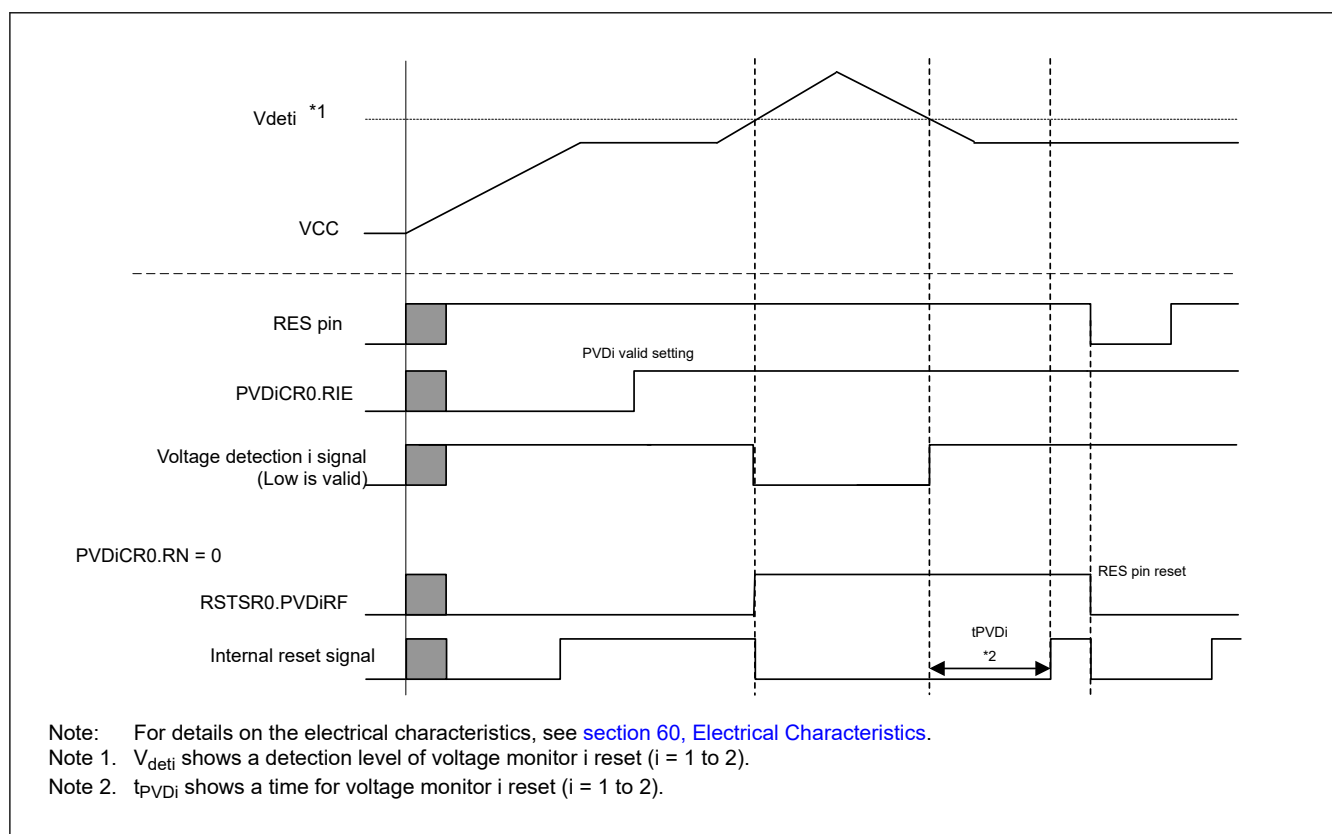


Figure 6.4 Example of operations during voltage monitor i reset (i = 1, 2)

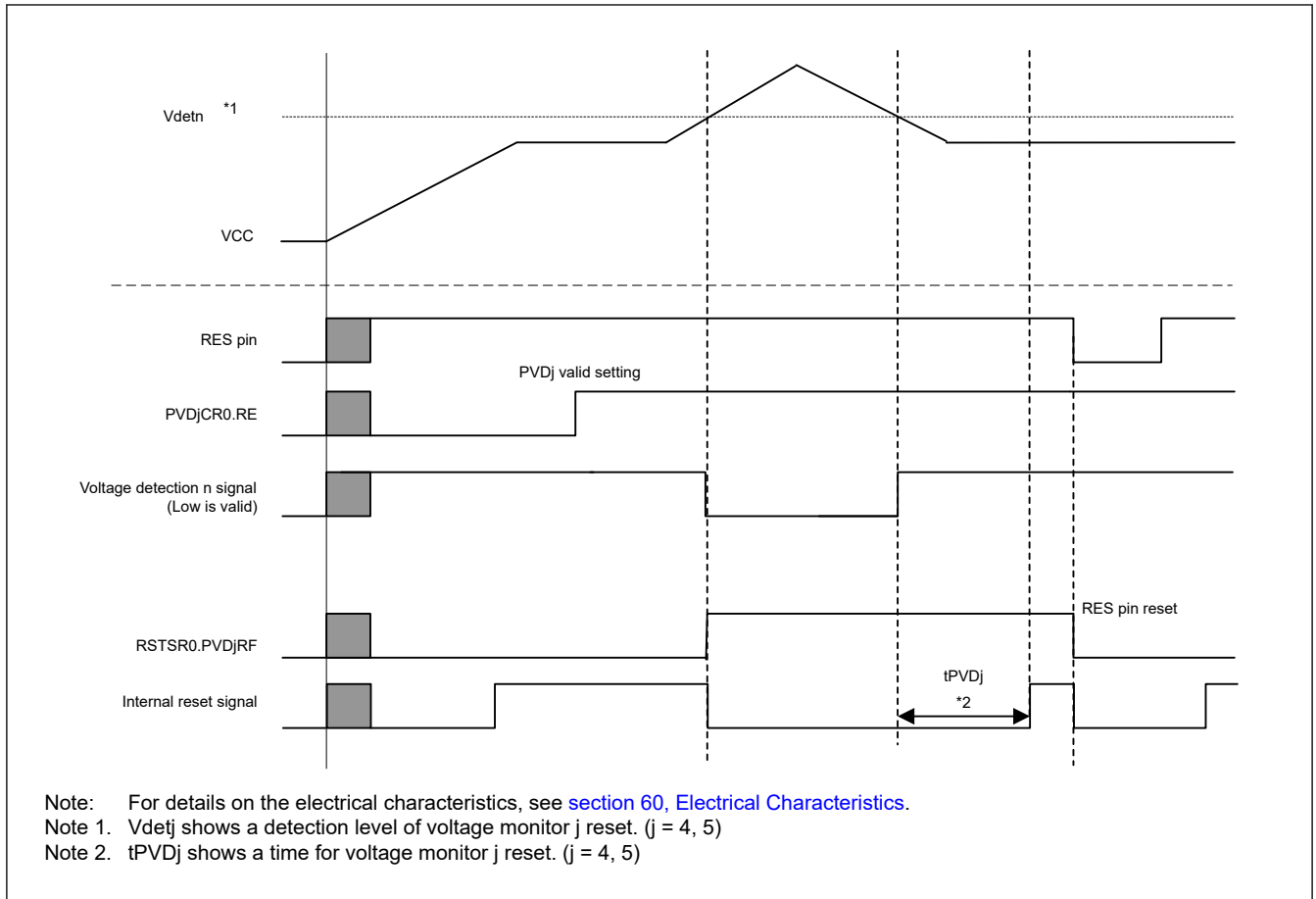


Figure 6.5 Example of operations during voltage monitor j reset ($j = 4, 5$)

The Core Voltage Monitor reset is an internal reset generated by the Core Voltage monitor circuit.

If the Core Voltage Detection Circuit disable bit (CVMRDIS) in Option Function Select Register 0 (OFS2) is 0 (Core Voltage monitor reset is enabled after a reset) and VDD falls below V_{det_VDDL} or rises above V_{det_VDDH} , the RSTR3.CVMRF flag is set to 1 and the Core Voltage detection circuit generates a Core Voltage monitor reset.

After VDD is between V_{det_VDDL} and V_{det_VDDH} and the Core Voltage monitor reset time (t_{CVM}) elapses, the internal reset is canceled, and the CPU starts the reset exception handling.

The Core Voltage Monitor reset function does not work during DSTBY mode.

When RES pin reset is asserted while issuing CVM reset, the CVM reset is not negated.

When the Core Voltage Monitor reset occurs, the values of register and SRAM cannot be guaranteed due to the response time until the reset occurs.

[Figure 6.6](#) shows examples of operations during the core voltage monitor reset.

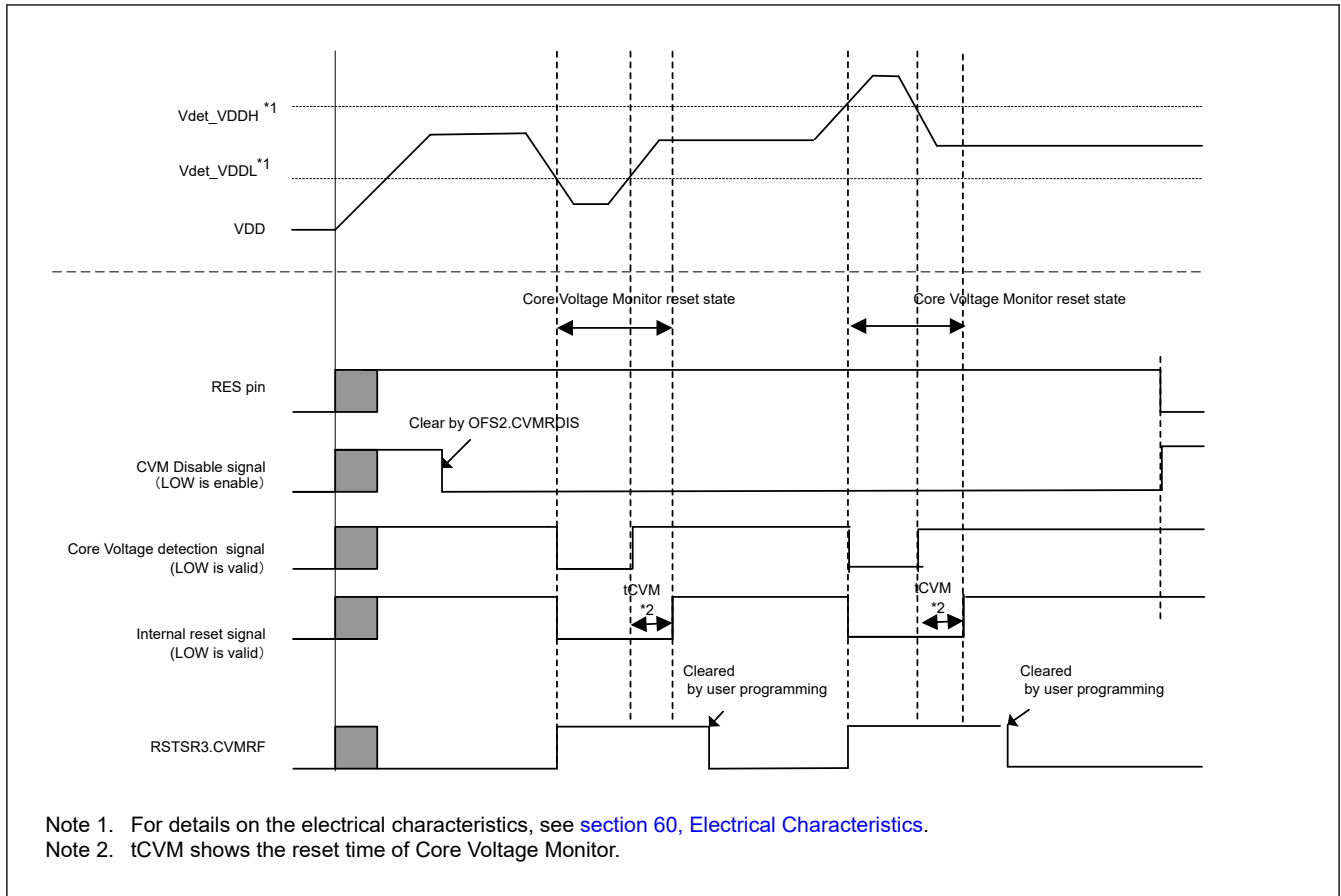


Figure 6.6 Operation Examples during Core Voltage Monitor Reset

6.3.4 Deep Software Standby Reset

These internal resets are generated when either of the Deep Software Standby mode are canceled by an associated interrupt. The Deep Software Standby reset is canceled after t_{DSBY} (return time after Deep Software Standby mode cancellation) elapses. At the same time, Deep Software Standby mode is also canceled.

When t_{DSBYWT} (wait time after Deep Software Standby mode cancellation) elapses after Deep Software Standby mode has been canceled, the internal reset is canceled, and the CPU starts the reset exception handling.

For details of the Deep Software Standby reset, see [section 11, Low Power Mode](#).

6.3.5 Independent Watchdog Timer Reset

The independent watchdog timer reset is an internal reset generated from the Independent Watchdog Timer (IWDT). Output of the reset from the IWDT can be selected in the Option Function Select Register 0 (OFS0).

When output of the independent watchdog timer reset is selected, the reset is generated if the IWDT underflows, or if data is written when refresh operation is disabled. When the internal reset time (t_{RESW2}) elapses after the independent watchdog timer reset is generated, the internal reset is canceled and the CPU starts the reset exception handling.

For details on the independent watchdog timer reset, see [section 28, Independent Watchdog Timer \(IWDT\)](#).

6.3.6 CPU0/CPU1 Watchdog Timer Reset

The CPU0/CPU1 watchdog timer reset is an internal reset generated from the Watchdog Timer (WDT0/1). Output of the reset from the WDT0/1 can be selected in the WDT Reset Control Register (WDTRCR) or Option Function Select register 0 (OFS0).

When output of the watchdog timer reset is selected, the reset is generated if the WDT underflows, or if data is written when refresh operation is disabled. When the internal reset time (t_{RESW2}) elapses after the watchdog timer reset is generated, the internal reset is canceled and the CPU starts the reset exception handling.

For details on the watchdog timer reset, see [section 27, Watchdog Timer \(WDT\)](#).

6.3.7 CPU0/CPU1 Lockup Reset

The CPU0 lockup reset is an internal reset generated by Arm core. Output of the reset from Arm core can be selected by CPU0LCKUPCR.OAD.

When output of the CPU0 lockup reset is selected, CPU0 generated if Arm core is in the lockup state. When the internal reset time (t_{RESW2}) elapses after the CPU0 lockup reset is generated, the internal reset is cancelled, and the CPU starts the reset exception handling.

Similarly, CPU1 has lockup reset. Output of the reset from Arm core can be selected by CPU1LCKUPCR.OAD.

For details on the CPU lockup reset, see [section 2, CPU](#).

6.3.8 Software Reset

This internal reset is generated by a software setting of the SYSRESETREQ bit in the AIRCR register in the Arm core. When the SYSRESETREQ bit is set to 1, a software reset is generated. When the internal reset time (t_{RESW2}) elapses after the software reset is generated, the internal reset is canceled, and the CPU starts the reset exception handling. For dual CPU systems, SYSRESETREQ of CPU0 will always generate a software reset but CPU1 reset request can be gated off by setting a control register accessible by secure software running on CPU0.

6.3.9 Bus Error Reset

Bus error reset is an internal reset generated by buses.

It is an integrated reset generated by buses, and it consists of the following:

- Bus error reset (MSAU error, MPU error, Illegal address error, Slave TrustZone Filter error, Slave Bus error, Bufferable write error)

Output of Bus error reset can be selected by MSAOAD.OAD or MPUOAD.OAD or BUSOAD.ILERRROAD or BUSOAD.SLERROAD or BUSOAD.BWERROAD.

When the internal reset time (t_{RESW2}) elapses after the Bus error reset is generated, the internal reset is cancelled, and the CPU starts the reset exception handling.

For details on the Bus error reset, see [section 15, Buses](#). Determination of reset generation source in Bus error reset is similar to determination of interrupt generation source, see [section 14, Interrupt Controller Unit \(ICU\)](#).

6.3.10 Common Memory Error Reset

Common memory error reset is an internal reset generated by SRAMs.

It is an integrated reset generated by SRAMs and consists of the following:

- SRAM error reset (ECC error)

Output of SRAM error reset can be selected by SRAMCRn.OAD.

When the internal reset time (t_{RESW2}) elapses after the Common memory error reset is generated, the internal reset is cancelled, and the CPU starts the reset exception handling.

For details on the Common memory error reset, see [section 57, SRAM](#). Determination of reset generation source in Common memory error reset is similar to determination of interrupt generation source, see [section 14, Interrupt Controller Unit \(ICU\)](#).

6.3.11 Local Memory 0 / Local Memory 1 Error Reset

Local memory 0/ Local memory 1 error reset is an internal reset generated by SRAMs.

It is an integrated reset generated by SRAMs, and it consists of the following:

- Local Memory 0 error reset (ECC error of CPU0 Cache or TCM)
- Local Memory 1 error reset (ECC error of CPU1 Cache or TCM)

Output of Local Memory 1 error reset can be selected by CAPOAD.OAD and TCMCRC.OAD or TCMCRS.OAD.

When the internal reset time (t_{RESW2}) elapses after the Local Memory 0/1 Error Reset is generated, the internal reset is cancelled, and the CPU starts the reset exception handling.

For details on the Local Memory 0 and Local Memory 1 error reset, see [section 2, CPU](#). For determination of reset generation source in Local Memory 0/Local Memory 1 Error Reset, it is similar to determination of interrupt generation source and see [section 14, Interrupt Controller Unit \(ICU\)](#).

6.3.12 Temperature Monitor Reset

Temperature monitor reset is an internal reset generated by temperature sensor (TSN). TSN detects ambient temperature on surrounding environment of MCU. If it is outside from the specified range, reset is issued.

The function of temperature monitor reset is enabled via specific procedure which is described in [Figure 6.7](#).

When TSN is activated and reset function enabled, temperature monitoring is executed. Representative behavior of temperature monitor reset is shown in [Figure 6.8](#) and [Figure 6.9](#).

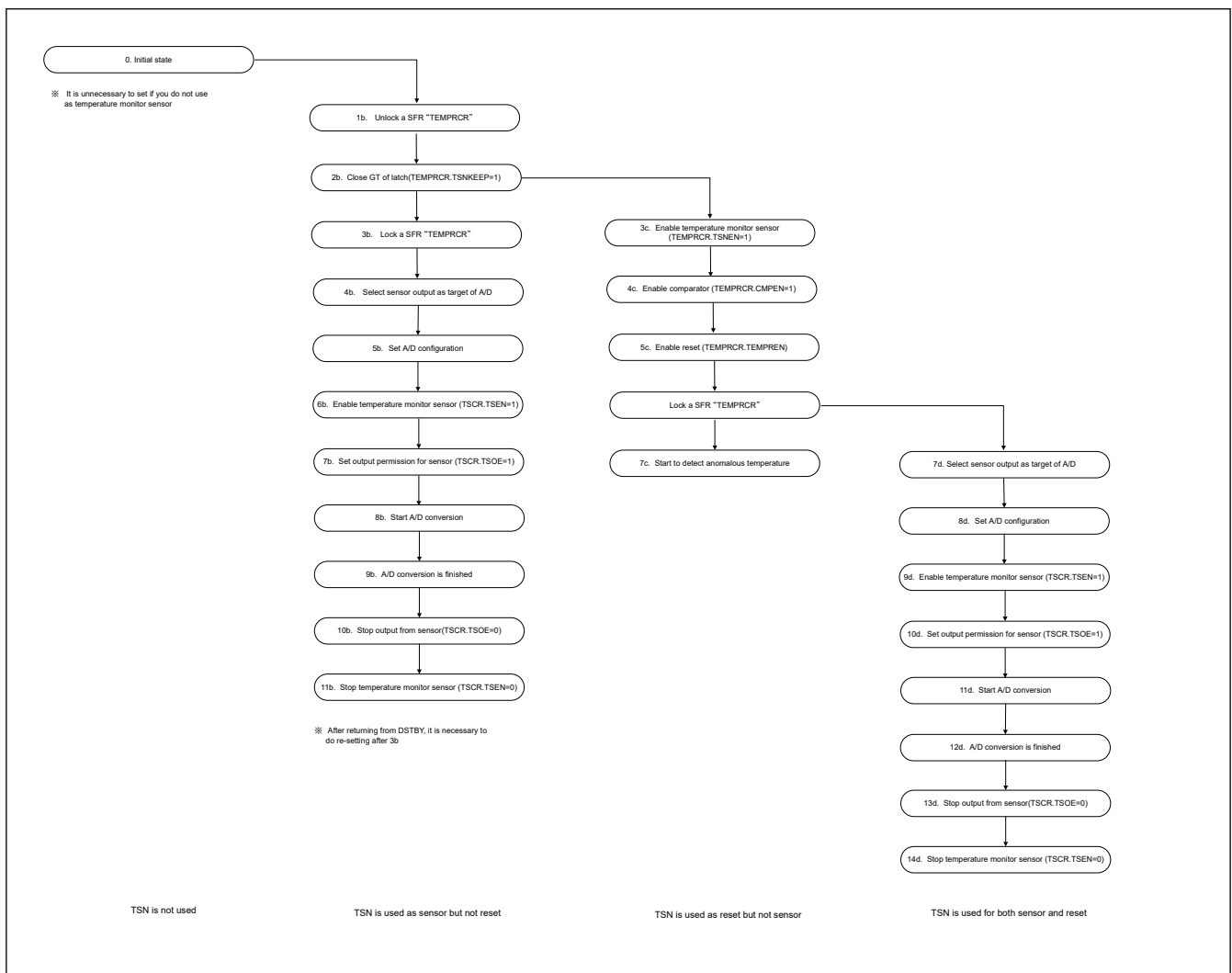


Figure 6.7 Setting procedure for temperature monitor sensor

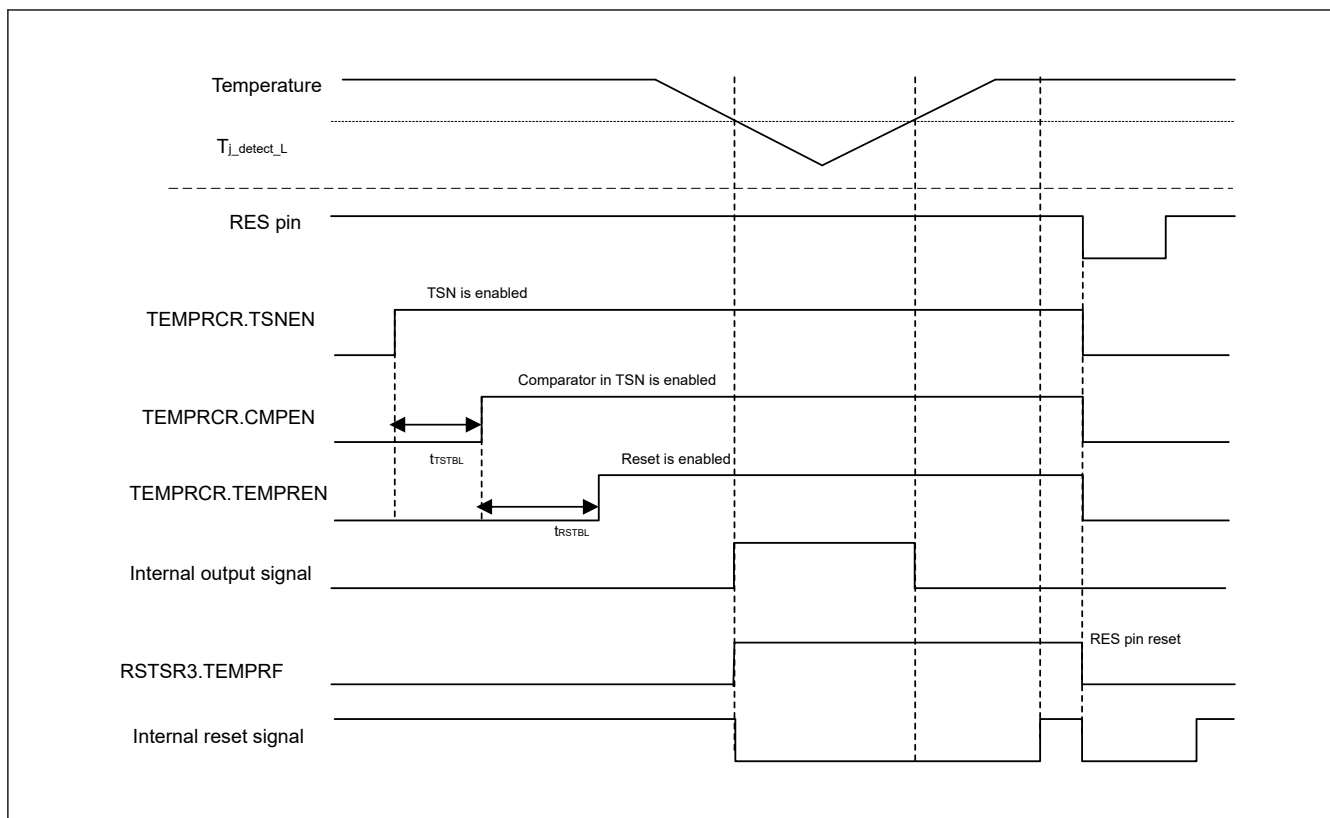


Figure 6.8 Example of temperature monitor reset operation (low detection)

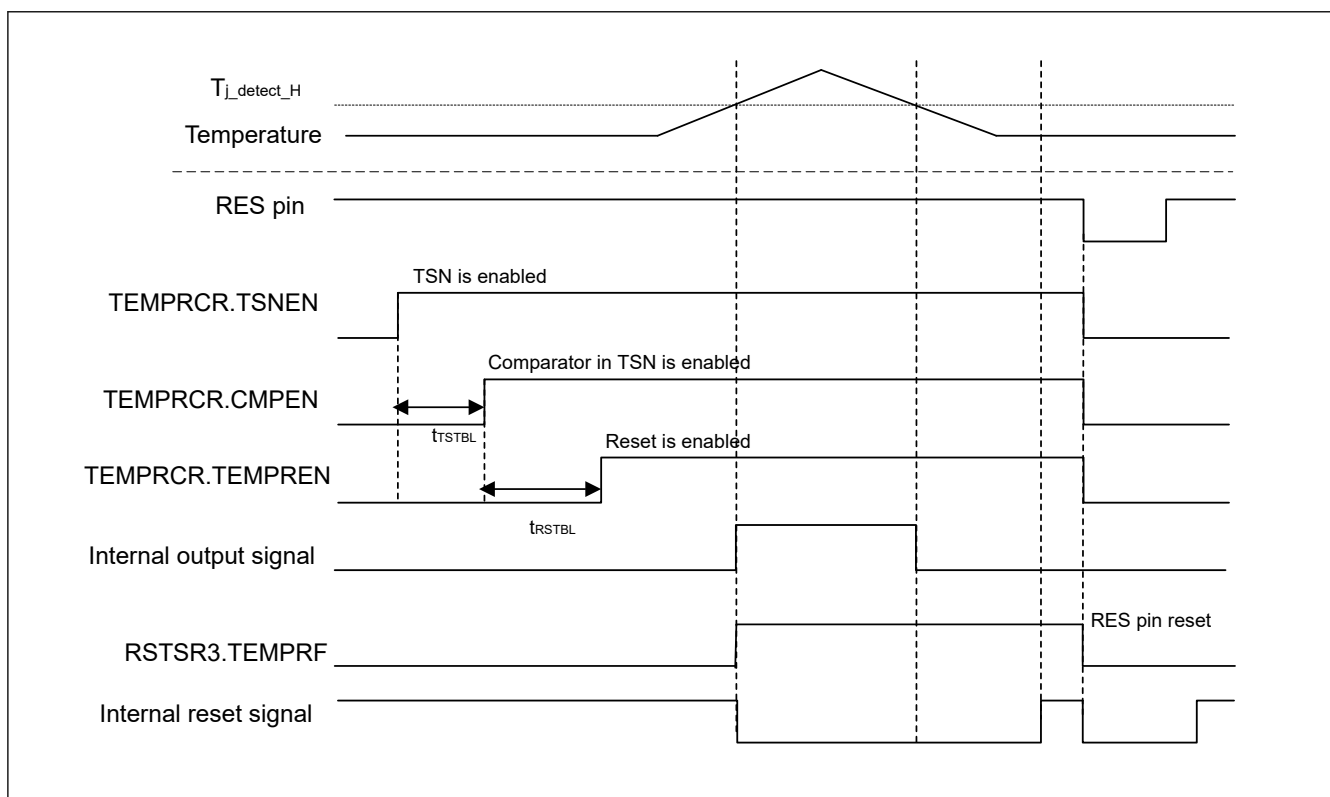


Figure 6.9 Example of temperature monitor reset operation (high detection)

6.3.13 Determination of Cold/Warm Start

Read the CWSF flag in RSTSR2 to determine the cause of reset processing. This flag indicates whether a power-on reset caused the reset processing (cold start) or a reset signal input during operation caused the reset processing (warm start).

The CWSF flag is set to 0 when a power-on reset occurs (cold start), otherwise the flag is not set to 0. The flag is set to 1 when 1 is written to it through software. It is not set to 0 even on writing 0 to it.

Figure 6.10 shows an example of cold/warm start determination operation.

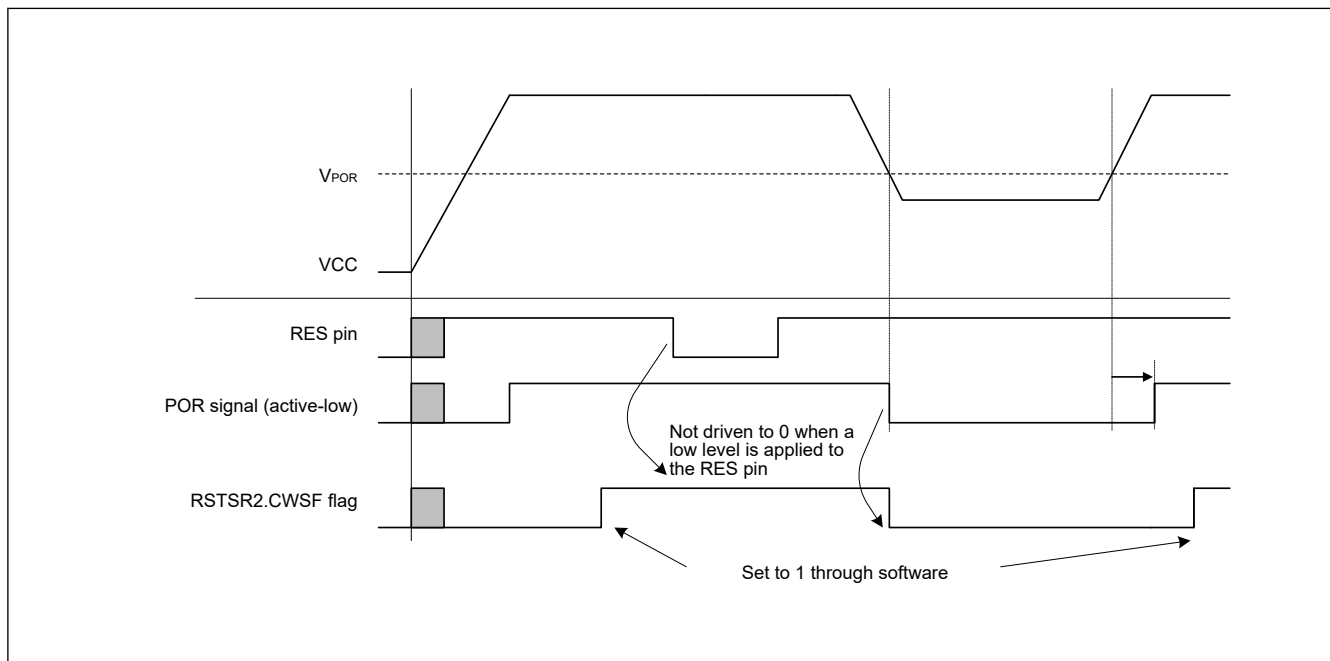


Figure 6.10 Example of cold/warm start determination operation

6.3.14 Determination of Reset Generation Source

Read RSTSR0, RSTSR1, and RSTSR3 to determine which reset executes the reset exception handling.

Figure 6.11 shows an example of the flow to identify a reset generation source. The reset flag must be written with 0 after 1 is read.

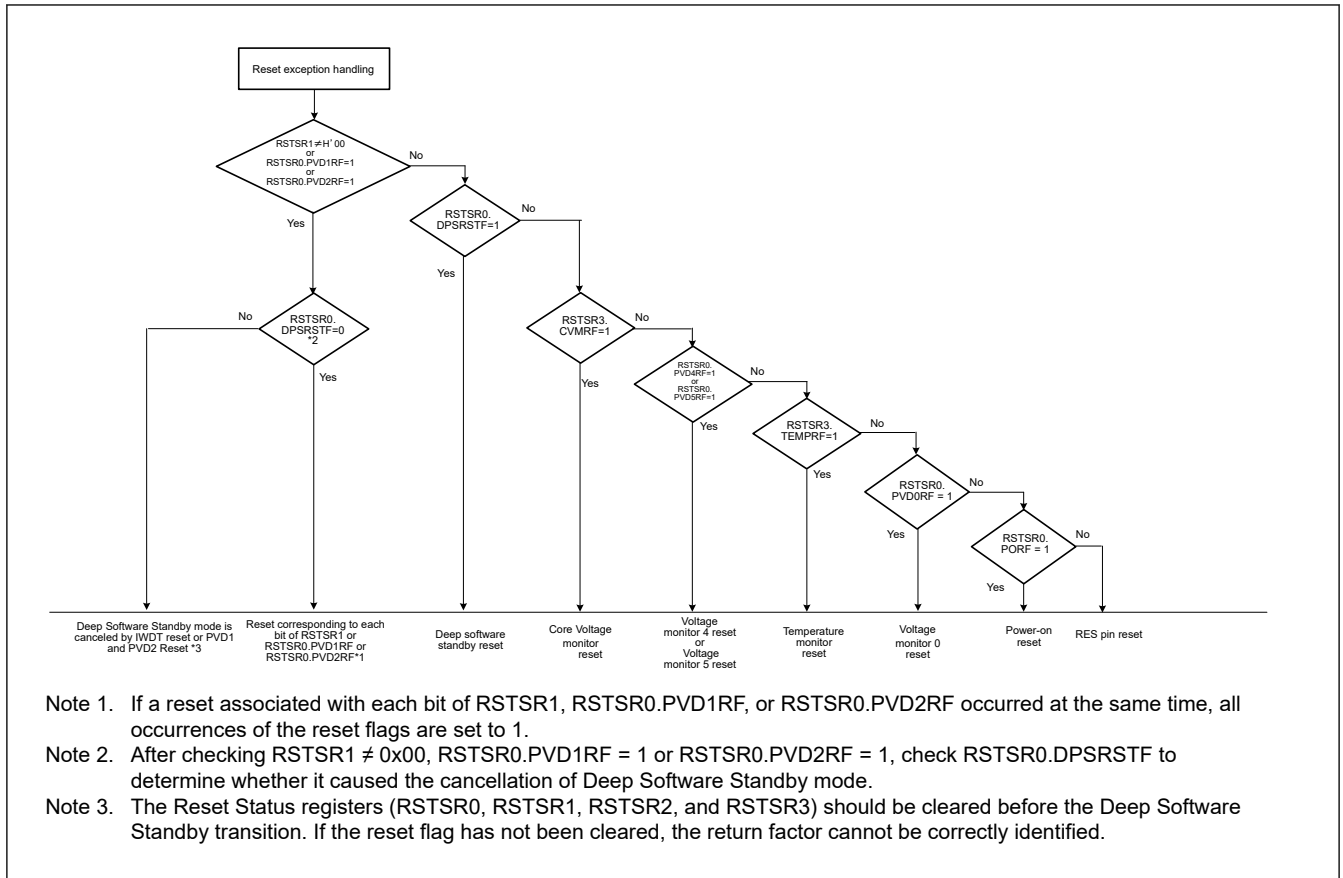


Figure 6.11 Example of reset generation source determination flow

6.4 Reset Protection from Non-Secure Programs

The reset occurrence can be controlled. A reset is generated according to the setting of SYRSTMSK0, SYRSTMSK1 and SYRSTMSK2 that can only be accessed from a secure program.

The reset source that is set to disable is generated but is not propagated to the MCU. The reset flags of RSTSR0 and RSTSR1 bits are not set.

The resets that can be controlled are as follows:

- Independent watchdog timer reset
- CPU0/CPU1 Watchdog timer reset
- CPU0/CPU1 lockup reset
- Common memory error reset
- Local memory 0 error reset
- Local memory 1 error reset
- Bus error reset
- Voltage monitor 1 reset
- Voltage monitor 2 reset
- Software reset